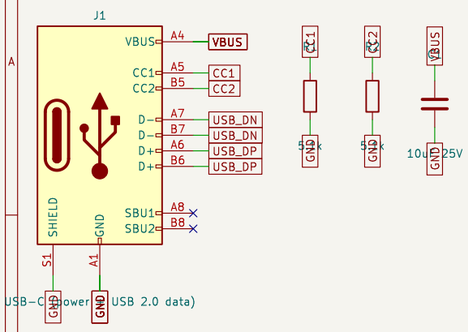
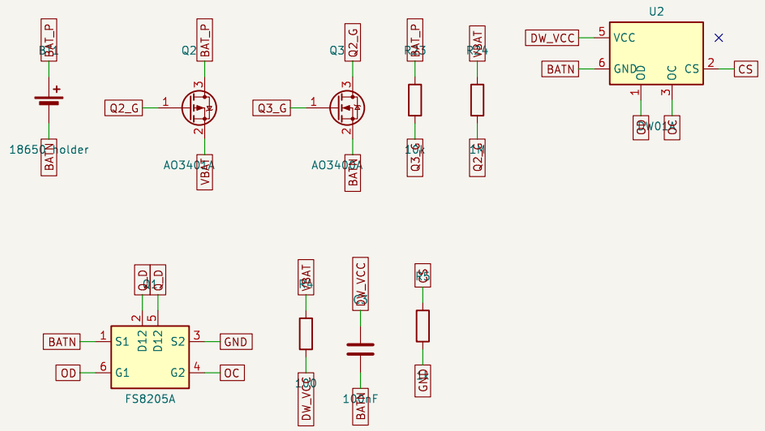
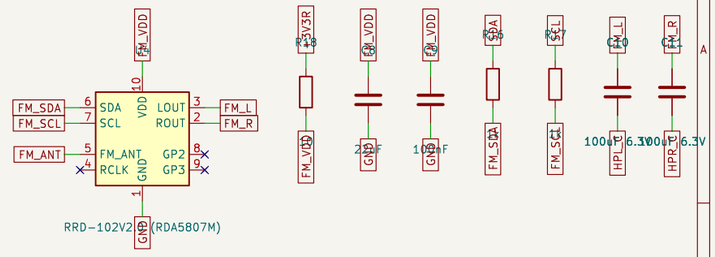
USB-C input (power + USB 2.0 data, CC 5.1k pull-downs)



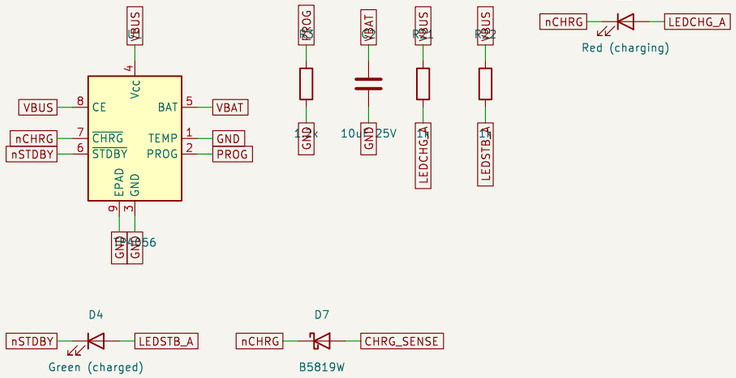
18650 + reverse-polarity FETs (Q2/Q3) + DW01A / FS8205A protection



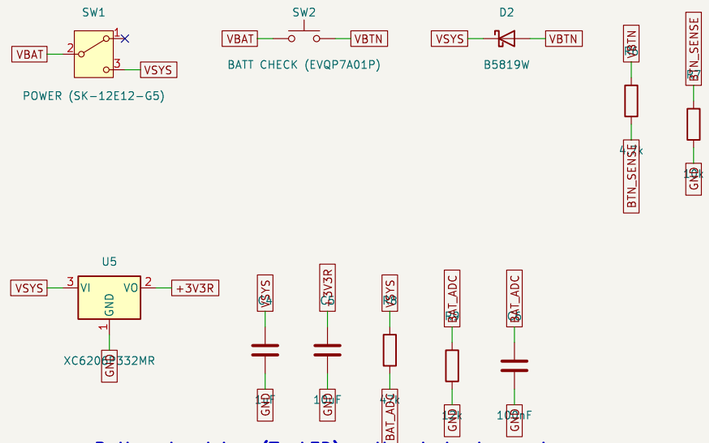
RRD-102 (RDA5807M) FM module, headphone out, headphone-ground antenna



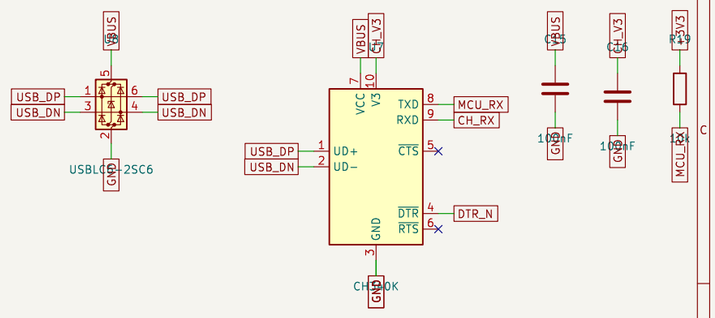
TP4056 Li-ion charger (~900 mA) + charge LEDs



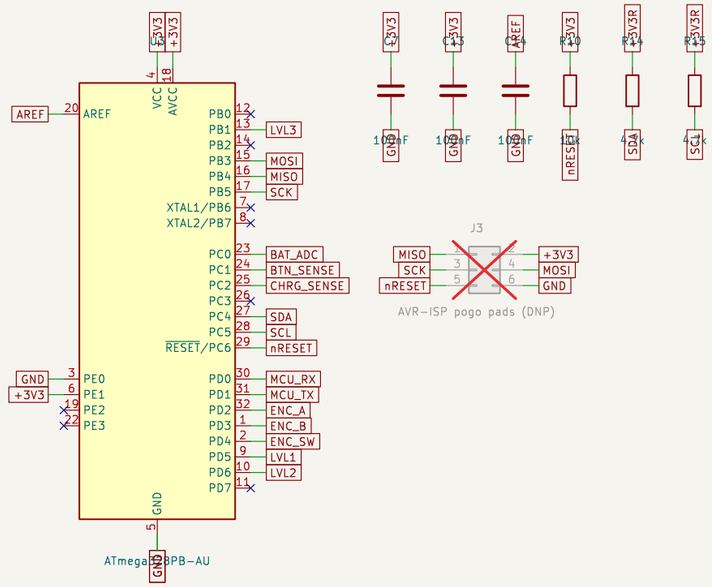
Power switch, battery-check button, radio 3.3 V LDO (+3V3R), battery ADC



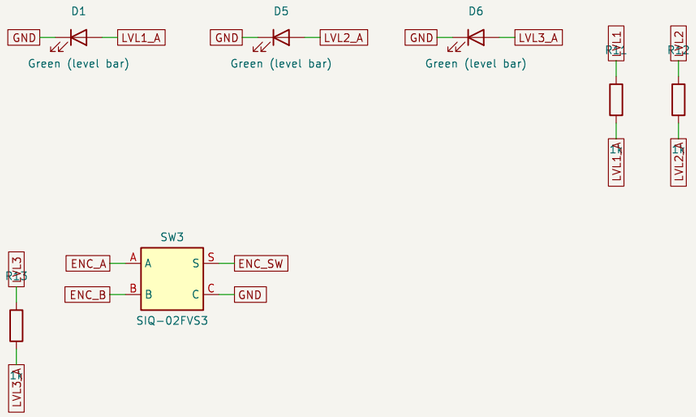
USB serial: CH340K (VBUS only) + ESD + auto-reset; MCU-only 3.3 V rail



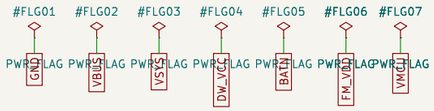
ATmega328PB MCU (internal 8 MHz) + AVR ISP pogo pads

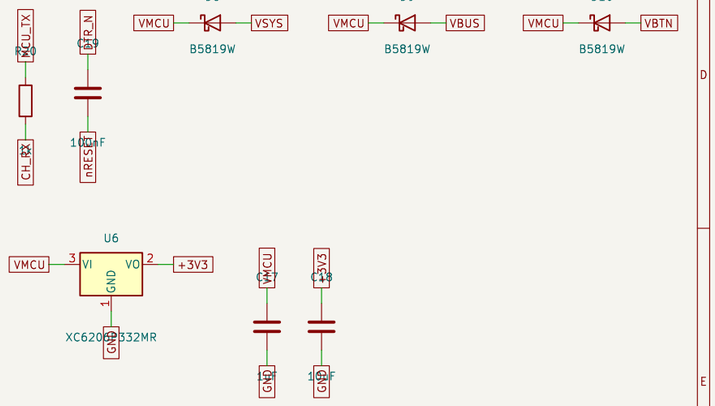


Battery level bar (3x LED) + thumbwheel encoder



Power flags (ERC)





ATmega328PB TQFP-32 (DS40001906): pin 3 PE0=GND, pin 6 PE1=+3V3 (328P-AU fits too)
Generated from scripts/design.py — every pin is labelled with its net name

Sheet: /
File: openald.kicad_sch

Title: OpenALD v1.0 — ATmega328PB + CH340K + RDA5807M, 18650, USB-C

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. 9.0.2		Id: 1/1